Experiment No 12

AIM: Study of Full wave rectifiers

APPARATUS: Rectifier Kit, Digital Multimeter, Connecting wires.

THEORY:

The circuit of a center-tapped full wave rectifier uses two diodes D1&D2. During positive half cycle of secondary voltage (input voltage), the diode D1 is forward biased and D2is reverse biased. So the diode D1 conducts and current flows through load resistor R_L.

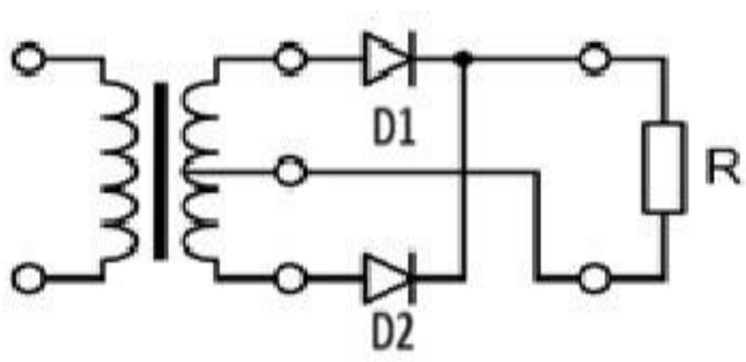
During negative half cycle, diode D2 becomes forward biased and D1 reversebiased. Now, D2 conducts and current flows through the load resistor R_L in the same direction. There is a continuous current flow through the load resistor R_L, during both the half cycles and will get unidirectional current as show in the model graph. The difference between full wave and half wave rectification is that a full wave rectifier allows unidirectional (one way) current to the load during the entire 360 degrees of the input signal and half-wave rectifier allows this only during one half cycle (180 degree).

CALCULATIONS:

Vrms = Vm/2 Vm =Vrms√2 Vdc=2Vm/Π

Ripple factor, $r = \sqrt{(Vrms/ Vdc)^2 - 1} = 0.48$

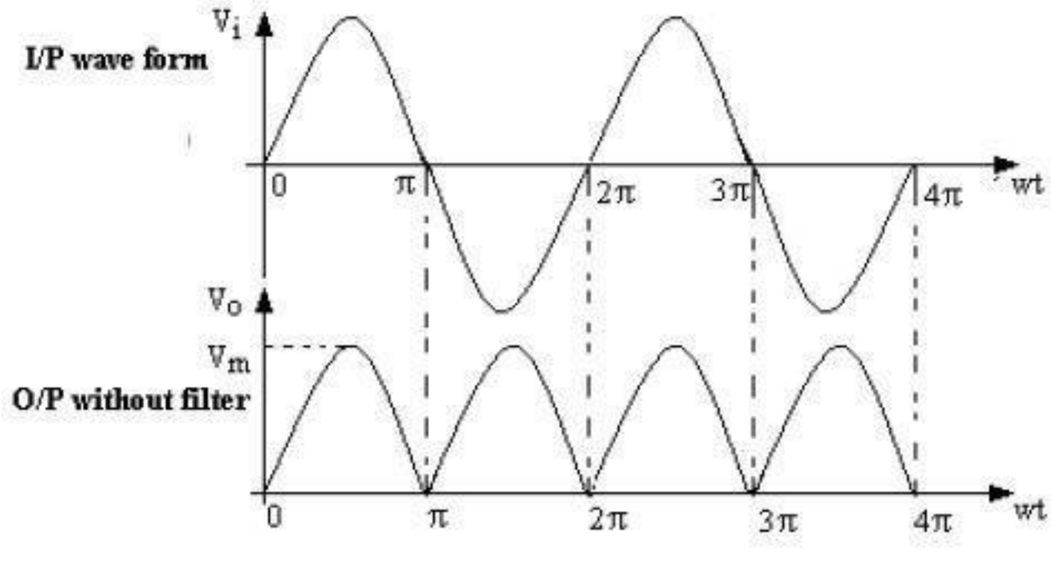
FULL WAVE RECTIFIER WITH FILTER:



Observation Table

Sr. No	Load Resister (RL)	vrms Value of AC signal		DC Voltage (Vdc)		Ripple Factor (γ)
		With C	Without C	With C	Without C	
1	47 Ω					
2	100 Ω					

WAVEFORMS:



PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Connect the ac mains to the primary side of the transformer and the secondaryside to the rectifier.
3. Measure the ac voltage at the input side of the rectifier.
4. Measure both ac and dc voltages at the output side the rectifier.
5. Find the theoretical value of the dc voltage by using the formula Vdc=2Vm/Π
6. Connect the filter capacitor across the load resistor and measure the values of Vacand Vdc at the output.
7. The theoretical values of Ripple factors with and without capacitor are calculated.
8. From the values of Vac and Vdc practical values of Ripple factors are calculated.The practical values are compared with theoretical values.

RESULT: Hence we are study of Full wave rectifier and find its ripple factor

PRECAUTIONS:

1. The primary and secondary side of the transformer should be carefully identified.
2. The polarities of all the diodes should be carefully identified.

VIVA QUESTIONS:

1. Define regulation of the full wave rectifier?
2. Define peak inverse voltage (PIV)? And write its value for Full-wave rectifier?
3. If one of the diode is changed in its polarities what wave form would you get?
4. Does the process of rectification alter the frequency of the waveform?
5. What is ripple factor of the Full-wave rectifier?
6. What is the necessity of the transformer in the rectifier circuit?
7. What are the applications of a rectifier?
8. What is meant by ripple and define Ripple factor?